HealthML

Sprint 4 Quality Assurance Review

Detailed re-test report with evidence and acceptance analysis

Project	HealthML
Branch	ai_implementation
Commit	770cef56d68304a4e7a96ea8e8b64b6e2116638e
QA Date	2026-04-15

Sprint 4 QA Review

Executive Summary

This report documents a fresh Sprint 4 QA re-test of the `ai_implementation` branch after the latest developer push. The tested commit is:

- Commit: `770cef56d68304a4e7a96ea8e8b64b6e2116638e`
- Commit date: `2026-04-14 22:08:40 +0300`
- Commit message: `Fix SMOTE overwriting original non-numeric column values`

The branch is materially improved and now passes the full pipeline and certificate requirements. However, two Sprint 4 acceptance items still fail in live execution:

1. Step 6 clinical sense-check text does not refresh correctly after switching domains and rebuilding the pipeline.

2. Step 7 subgroup table still omits the required age-band rows.

Because both are explicitly required by the Sprint 4 brief, the final verdict remains:

Final Verdict: Conditional Fail

The branch is close to Sprint 4 acceptance and is demo-capable, but it is not yet safe to present as fully complete against the stated Sprint 4 deliverables.

Branch Integrity Note

- Tested branch: `origin/ai_implementation`
- Tested worktree: `/tmp/HealthML-ai_implementation_latest3`
- Tested commit: `770cef56d68304a4e7a96ea8e8b64b6e2116638e`
- Integrity note: all QA execution in this report was performed against that exact commit after a fresh fetch and clean worktree creation.
- Branch model set note: this branch includes `Naive Bayes` as the sixth model; it does **not** expose `Neural Network`.

Sprint 4 Spec Basis

This QA cycle was executed directly against the Sprint 4 brief provided by the team. The most relevant acceptance items were:

- `Clinical Sense-Check Banner`: domain-specific text implemented for all 20 domains; correct text appears when switching domains.
- `Step 7 — Subgroup Table`: rows for male/female/age groups.
- `Bias Auto-Detection Banner`: triggered when subgroup sensitivity is more than 10 percentage points below overall.
- `EU AI Act Checklist`: 8 items, 2 pre-checked, visible completion progress.
- `PDF Certificate Generation`: backend endpoint working and tested for at least 3 different domains.
- `Full Pipeline Test Report`: end-to-end Steps 1–7 with fresh CSV and all acceptance criteria verified.

This matters because the two remaining defects are not interpretive preferences. They are direct mismatches against explicit Sprint 4 acceptance text.

Test Environment

Item	Value
QA execution timestamp	`2026-04-15T12:47:27+03:00`
OS	Linux sandbox / local dev runtime
Frontend	Vite dev server on `http://127.0.0.1:5173`

Backend	Express on `http://127.0.0.1:3001`
ML service	FastAPI on `http://127.0.0.1:8000`
Backend health	`{"status":"ok","message":"ML Tool Backend is running"}`
ML health	`{"status":"ok","message":"HealthML Python ML service running"}`
Fresh CSV test data	`tests/fixtures/fresh_patients.csv`

Test Methodology

This QA review used two complementary test tracks:

1. The branch's own Playwright acceptance suite.

2. An independent Sprint 4 re-test suite written specifically to verify the brief and produce audit-friendly evidence.

Execution included:

- live frontend navigation
- fresh CSV upload
- column mapping and schema validation
- data preparation
- model training
- Step 6 and Step 7 validation
- backend PDF certificate validation
- route alias validation
- source-level verification for threshold and domain-map rules

Execution Log

Run A — Branch Acceptance Suite

- Command:

```
npx playwright test tests/e2e/sprint4_acceptance.spec.ts --reporter=list,html
```

- Result: `15 passed / 18 total`
- Exit code: `1`

Observed failures in the branch-owned suite:

- certificate test payload is stale relative to the current backend request contract and returns `422`
- home-page test assumes `Logistic Regression` text is visible before navigating into the model-selection step
- one source assertion expects the old fairness label glyph instead of the current `[!] Review Needed`

Assessment: these three failures are mainly test-suite drift, not the primary product blockers found in this review.

Run B — Independent Sprint 4 Re-Test Suite

- Command family:

```
npx playwright test tests/e2e/sprint4_retest.spec.ts --reporter=list,html
npx playwright test tests/e2e/sprint4_retest.spec.ts --grep 'RT-04|RT-05|RT-06|RT-07|RT-08|RT-09|RT-10|RT-11' --reporter=list,html
npx playwright test tests/e2e/sprint4_retest.spec.ts --grep 'RT-06|RT-07|RT-08|RT-09|RT-10|RT-11' --reporter=list,html
```

- Aggregated independent result: `9 passed / 11 total`

Independent Re-Test Results

Test ID	Scope	Result	Notes
RT-01	Full Steps 1–7 flow with fresh CSV	Pass	No crash; fresh CSV flow completed end-to-end
RT-02	Step 6 explainability UI	Pass	3-patient selector, clinical labels, red/green contribution bars visible
RT-03	Domain-switch sense-check refresh	Fail	Clinical sense-check text stayed cardiology-specific after domain switch and pipeline rebuild
RT-04	Step 7 subgroup table	Fail	`Male` and `Female` rows

			visible, but age-band rows were missing
RT-05	Step 7 case-study color structure	Pass	Red / amber / green case-study cards rendered
RT-06	`/api/ml/generate-certificate`	Pass	PDF returned successfully; generation well below 10s
RT-07	`/api/generate-certificate` alias route	Pass	Exact Sprint 4 route contract is now satisfied
RT-08	Certificate content for 3 domains	Pass	Domain and model values were correctly rendered in PDF output
RT-09	Fairness threshold source rule	Pass	`if gap > 0.10` present in `backend/ml/main.py`
RT-10	Representation threshold source rule	Pass	`maxRepGap > 15` present in `EthicsBias.tsx`
RT-11	20-domain mapping + model set	Pass	20 domain keys present; branch model set includes `Naive Bayes`

Release-Blocking Defects

Defect ID	Severity	Sprint 4 Item	Description	Evidence	Recommended Owner
RB-01	High	Clinical Sense-Check Banner	After switching from `cardiology_heart_failure` to `radiology_pneumonia`, the Step 6 amber banner still displayed the cardiology-specific sense-check text. The banner did not refresh to the new domain.	RT-03 failure, screenshot evidence, live re-run	Frontend / state-management owner
RB-02	High	Step 7 Subgroup Table	Live Step 7 output showed `Male` and `Female` rows, but no `Age 18-60`, `Age 61-75`, or `Age 76+` rows. The Sprint 4 brief explicitly requires male/female/age-group rows.	RT-04 failure, Step 7 screenshot evidence	ML service / fairness-data owner

Detailed Findings

RB-01 — Step 6 banner does not update after domain switch

What was tested

- Complete pipeline was run with one specialty.
- Step 6 banner text was captured.
- The workflow then switched specialty using the reset confirmation flow.
- The pipeline was rebuilt from Step 2 onward.
- Step 6 banner text was captured again.

Observed result

- The second Step 6 banner remained cardiology-specific.
- Expected radiology-specific text did not appear.

Why this matters

Sprint 4 explicitly requires that the clinical sense-check banner changes correctly when switching domains. This is one of the instructor spotlight checks in the showcase agenda.

Likely cause

The `SPECIALTY_SENSE` map exists in source and contains all 20 domains, so the issue is likely not missing content. The defect is more likely caused by stale specialty state, incomplete reset propagation, or a mismatch between the specialty stored in context and the value used at Step 6 render time.

Relevant code:

- `app/components/steps/Explainability.tsx`
- `app/components/MLContext.tsx`
- `app/components/steps/ClinicalContext.tsx`

RB-02 — Step 7 subgroup table is missing age-band rows

What was tested

- Fresh CSV flow was executed through Step 7.
- The subgroup performance table was inspected for:
- `Male`
- `Female`
- age-band rows

Observed result

- `Male` row present
- `Female` row present
- no age-band row visible

Why this matters

Sprint 4 explicitly requires subgroup rows for:

- male
- female
- age groups

The current live output therefore does not fully satisfy the Step 7 table deliverable.

Technical note

`backend/ml/main.py` contains age-band generation logic:

- `Age 18-60`
- `Age 61-75`
- `Age 76+`

So this is not a missing-source issue. It is a runtime-output issue. The most likely causes are:

- band population falling below the runtime `mask.sum() < 5` threshold
- fairness pipeline not returning age rows in the live scenario used by the frontend
- age column transformation side-effects during train / fairness stages

Relevant code:

- `backend/ml/main.py`
- `app/components/steps/EthicsBias.tsx`

Non-Blocking Observations

Observation ID	Severity	Notes
O-01	Medium	The branch's own acceptance suite still has 3 failures, but they are mostly test-suite drift rather than primary product regressions.
O-02	Low	Step 7 case-study cards are structurally correct in red/amber/green, but their headings are descriptive scenario titles rather than the literal brief words `failure`, `near-miss`, and `prevention`. This is not currently treated as a blocker.
O-03	Low	Backend health message still says `ML Tool Backend` instead of `HealthML`; branding consistency can be cleaned up.

Evidence Register

Evidence ID	Description	File
E-01	Full pipeline completion at Step 7	[rt01-step7-complete.png](/tmp/HealthML-ai_implementation_latest3/qa-evidence/rt01-step7-complete.png)
E-02	Step 6 explainability with 3-patient selector and waterfall bars	[rt02-step6-explainability.png](/tmp/HealthML-ai_implementation_latest3/qa-evidence/rt02-step6-explainability.png)
E-03	Initial Step 6 amber/blue banner state before domain switch	[rt03-step6-banners-domain1.png](/tmp/HealthML-ai_implementation_latest3/qa-evidence/rt03-step6-banners-

		domain1.png)
E-04	Step 7 case-study cards section	[rt05-step7-case-studies.png](/tmp/HealthML-ai_implementation_latest3/qa-evidence/rt05-step7-case-studies.png)
E-05	Playwright HTML report	[index.html](/tmp/HealthML-ai_implementation_latest3/playwright-report/index.html)

Risk Assessment

Risk ID	Risk	Impact	Status
R-01	Domain banner defect remains during showcase	Instructor can immediately detect wrong clinical explanation when switching domains	Open
R-02	Missing age-band rows remain in subgroup table	Step 7 deliverable can be judged incomplete	Open
R-03	Team-owned acceptance suite remains partially stale	Automated evidence may appear weaker than actual product state	Open

Remediation Plan

Action ID	Action	Priority	Owner	Due Date
A-01	Fix specialty-change propagation so Step 6 banner always reflects the active domain after reset and retrain	P1	Frontend owner	Before next demo / retest
A-02	Trace fairness payload generation and ensure age-band rows are returned and rendered in Step 7	P1	ML service owner	Before next demo / retest
A-03	Update branch-owned Playwright suite to current UI and certificate payload contract	P2	QA / automation owner	After blocker fixes

Latest Revalidation Note

Latest revalidation on `2026-04-15` confirms the same high-level picture:

- UI pipeline is now functional end-to-end
- certificate generation is fixed
- exact `/api/generate-certificate` route is fixed
- Sprint 4 blocker status now depends on two remaining live UI/data mismatches:
- domain sense-check refresh
- age-band subgroup rows

Conclusion

This branch is substantially stronger than prior QA cycles. The most difficult Sprint 4 backend requirements are now satisfied:

- full 7-step pipeline works
- backend certificate generation works
- exact route alias works
- certificate content validation works for three domains
- fairness and representation thresholds are aligned in source

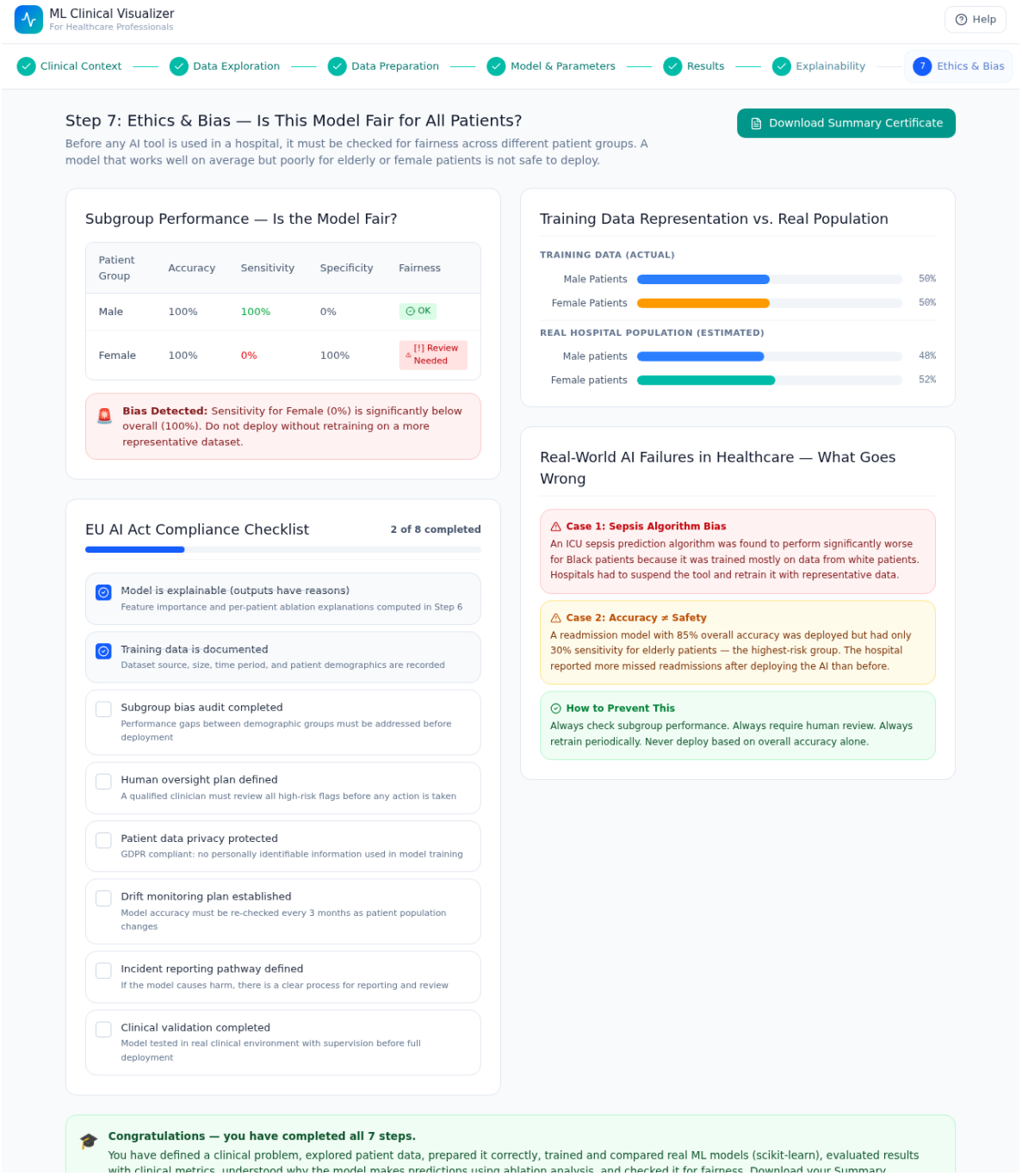
However, Sprint 4 is still not fully complete because two explicit acceptance criteria remain unmet in live execution.

QA Decision

- Demo readiness: `Partial / strong`
- Sprint 4 acceptance readiness: `Conditional Fail`
- Safe claim: `Do not present as fully complete until RB-01 and RB-02 are fixed and re-tested`

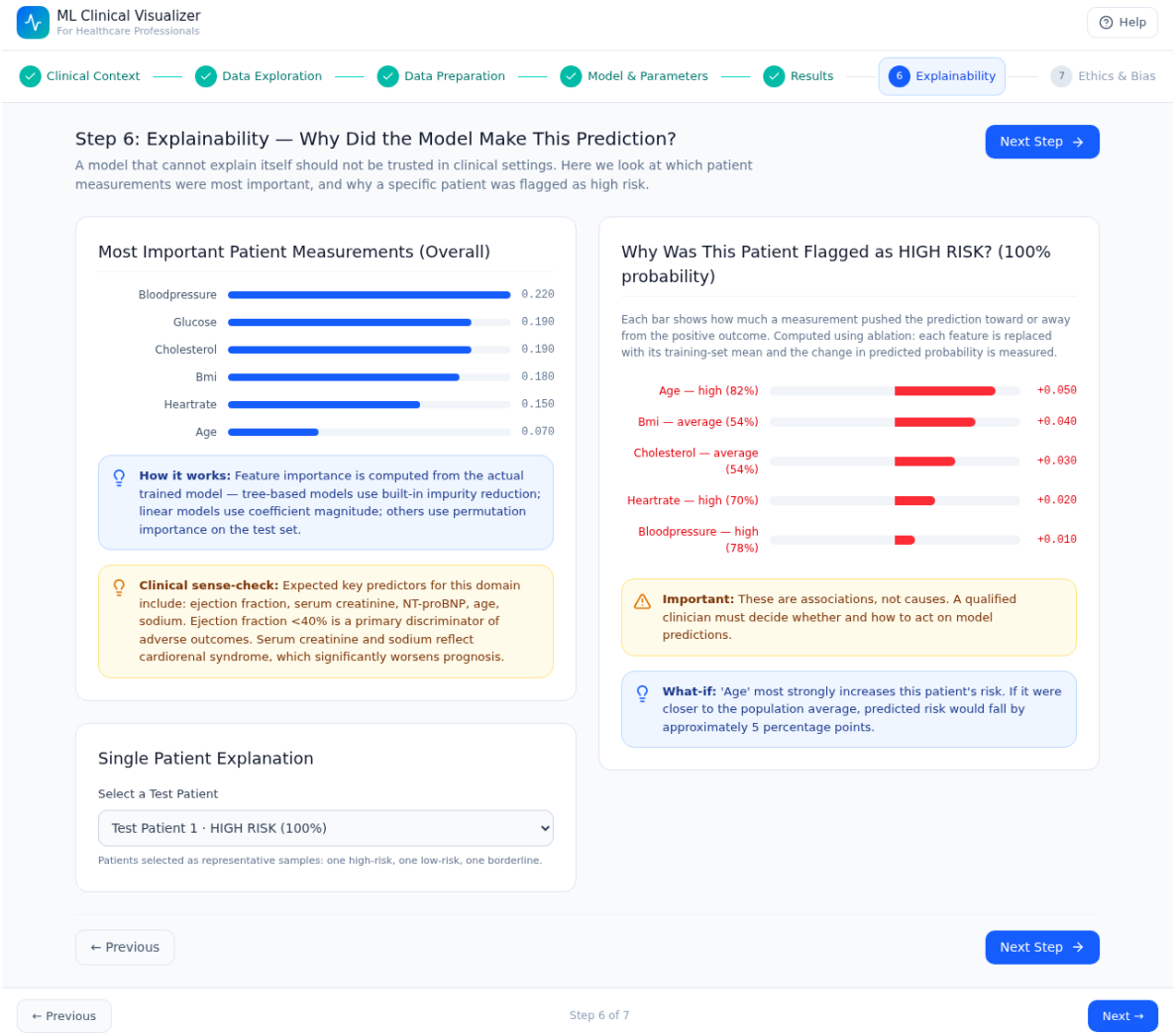
Embedded Screenshot Appendix

E-01 Full Pipeline



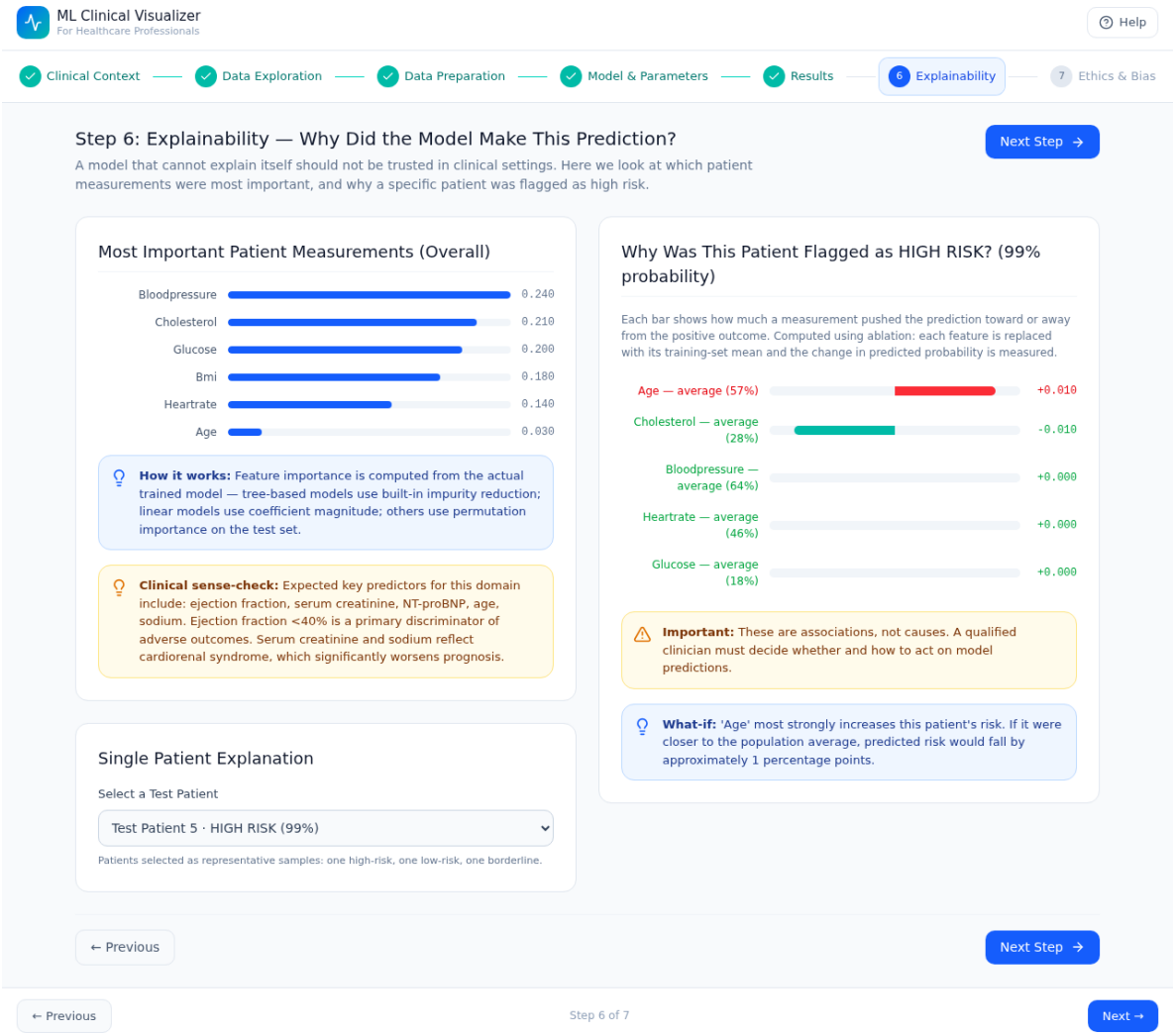
Full Pipeline Step 7

E-02 Step 6 Explainability



Step 6 Explainability

E-03 Step 6 Banner Before Domain Switch



Step 6 Banner Domain 1

E-04 Step 7 Case Study Cards

ML Clinical Visualizer
For Healthcare Professionals

Help

Clinical Context — Data Exploration — Data Preparation — Model & Parameters — Results — Explainability — 7 Ethics & Bias

Step 7: Ethics & Bias — Is This Model Fair for All Patients?

Download Summary Certificate

Before any AI tool is used in a hospital, it must be checked for fairness across different patient groups. A model that works well on average but poorly for elderly or female patients is not safe to deploy.

Subgroup Performance — Is the Model Fair?

Patient Group	Accuracy	Sensitivity	Specificity	Fairness
Male	100%	100%	0%	OK
Female	100%	0%	100%	[!] Review Needed

Bias Detected: Sensitivity for Female (0%) is significantly below overall (100%). Do not deploy without retraining on a more representative dataset.

Training Data Representation vs. Real Population

TRAINING DATA (ACTUAL)

Male Patients 50%
Female Patients 50%

REAL HOSPITAL POPULATION (ESTIMATED)

Male patients 48%
Female patients 52%

EU AI Act Compliance Checklist

2 of 8 completed

- [x] Model is explainable (outputs have reasons)
  Feature importance and per-patient ablation explanations computed in Step 6
- [x] Training data is documented
  Dataset source, size, time period, and patient demographics are recorded
- [ ] Subgroup bias audit completed
  Performance gaps between demographic groups must be addressed before deployment
- [ ] Human oversight plan defined
  A qualified clinician must review all high-risk flags before any action is taken
- [ ] Patient data privacy protected
  GDPR compliant: no personally identifiable information used in model training
- [ ] Drift monitoring plan established
  Model accuracy must be re-checked every 3 months as patient population changes
- [ ] Incident reporting pathway defined
  If the model causes harm, there is a clear process for reporting and review
- [ ] Clinical validation completed
  Model tested in real clinical environment with supervision before full deployment

Real-World AI Failures in Healthcare — What Goes Wrong

Case 1: Sepsis Algorithm Bias
An ICU sepsis prediction algorithm was found to perform significantly worse for Black patients because it was trained mostly on data from white patients. Hospitals had to suspend the tool and retrain it with representative data.

Case 2: Accuracy ≠ Safety
A readmission model with 85% overall accuracy was deployed but had only 30% sensitivity for elderly patients — the highest-risk group. The hospital reported more missed readmissions after deploying the AI than before.

How to Prevent This
Always check subgroup performance. Always require human review. Always retrain periodically. Never deploy based on overall accuracy alone.

Congratulations — you have completed all 7 steps.
You have defined a clinical problem, explored patient data, prepared it correctly, trained and compared real ML models (scikit-learn), evaluated results with clinical metrics, understood why the model makes predictions using ablation analysis, and checked it for fairness. Download your Summary Certificate to document what you built.

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Step 7 of 7

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Step 7 Case Study Cards